



SESSION 8 | COURSE 5140602

Migration and Urbanization in a Connected World

Fundamental Demographic Analysis | College of Population Studies, Chulalongkorn University

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By the End of This Session, You Will Be Able To...



Define migration, emigration, immigration, and change of residence precisely, and identify which side of a move each term describes.



Compute and interpret gross/net migration, and emigration/immigration/net migration rates from real data.



Explain migration selectivity and “brain drain” with current Thai and global evidence.



Compare traditional data sources (census, registers, surveys) with digital trace data, and name each one's blind spot.



Describe how mobile-phone, satellite, and social-media data support near-real-time mobility and displacement estimation.



Analyze current global and Thai urbanization trends, their drivers, and how “urban” is measured and redefined.

Agenda — 180 Minutes



BLOCK 1

Foundations — defining and measuring migration



BREAK

15-minute break



BLOCK 2

Selectivity, brain drain & traditional vs. digital data sources



BLOCK 3

Urbanization in a connected world, group work & practice

A World on the Move

304M

international migrants worldwide by 2024 — 3.7% of humanity, and still rising.

45%

of the global population already lives in cities as of 2025 — more than double the 1950 share.

83.4M

people living in internal displacement at end-2024 — a record, driving demand for faster, digital ways to track mobility.

Sources: IOM (2024), World Migration Report 2024; UN DESA (2025), World Urbanization Prospects 2025; IDMC (2025), Global Report on Internal Displacement.



BLOCK 1

Defining and Measuring Migration

The vocabulary and formulas demographers use to describe who moves, where, and how much.

What Counts as “Migration”?



Migration

A change of usual residence across a defined boundary (a district, region, or international border), typically for a minimum duration (e.g., ≥ 1 year).



Change of Residence

The broader event — moving one's usual dwelling — of which migration is the subset that crosses a specified boundary.



Emigration

Departure from one's country of usual residence to settle (temporarily or permanently) elsewhere.



Immigration

Arrival into a country to take up usual residence, from the perspective of the receiving country.

Emigration and Immigration Are Two Ends of One Move



ORIGIN COUNTRY

This move counts as EMIGRATION here — a departure recorded by the sending country's statistical system.



DESTINATION COUNTRY

This move counts as IMMIGRATION here — an arrival recorded by the receiving country's statistical system.

The same event is simultaneously one country's emigration and another's immigration — global totals should, in principle, match, but in practice rarely do, due to differing definitions and data lags.

Migration Streams and Counterstreams



Stream (A → B)

The larger, dominant flow of migrants moving from origin area A to destination area B in a given period.



Counterstream (B → A)

The smaller, offsetting flow moving in the opposite direction, from B back to A, in the same period.

Net migration alone can mask large, offsetting two-way flows — studying streams and counterstreams separately reveals dynamics that net figures alone hide.

Try It: Compute Gross Interchange

A researcher observes a migration stream from Province A to Province B of 12,000 people, and a smaller counterstream from B to A of 4,500 people in the same year.

STREAM (A→B)

12,000

12,000

COUNTERSTREAM (B→A)

4,500

4,500

GROSS INTERCHANGE

12,000 + 4,500

16,500

DISCUSS: Net migration between A and B here is only $12,000 - 4,500 = 7,500$ — far smaller than the 16,500 total volume of movement. Why might policymakers care about both figures?

Six Things Migration Statistics Are Not



“Migrant” Is Not Universal

Countries define “migrant” using different minimum-duration thresholds (3 months, 1 year, 5 years) — figures are often not directly comparable.



Net ≠ Gross

A net migration rate near zero can hide enormous two-way gross flows — never assume “no net change” means “no movement.”



Not Random

Migrants are selected on age, education, and skill — treating a migrant stream as a random sample of the origin population is a modeling error.



Not Always Voluntary

Migration statistics blend voluntary labor migration with forced displacement; conflating the two obscures very different policy needs.



Registers ≠ Reality

Administrative registers only capture what is legally registered — irregular and undocumented migration is systematically undercounted.



Emigration ≠ Immigration Totals

Because definitions and timing differ, world emigration and immigration totals rarely match exactly, even though every move is both.

Gross Migration vs. Net Migration



GROSS MIGRATION

Gross migration = in-migrants + out-migrants (internal), or immigrants + emigrants (international).

Captures the total volume of movement, regardless of direction.



NET MIGRATION

Net migration = in-migrants (immigrants) – out-migrants (emigrants).

Positive = net gain (net immigration); negative = net loss (net emigration).

Migration Rates

Emigration rate

$$(\text{Emigrants} \div \text{mid-year population of origin}) \times 1,000$$

Immigration rate

$$(\text{Immigrants} \div \text{mid-year population of destination}) \times 1,000$$

Net migration rate

$$[(\text{Immigrants} - \text{Emigrants}) \div \text{mid-year population}] \times 1,000$$

Gross migration rate

$$[(\text{Immigrants} + \text{Emigrants}) \div \text{mid-year population}] \times 1,000$$

Note: unlike vital rates, migration rates require a clear reference period and boundary definition — comparability across countries is limited by differing definitions of “migrant.”

Three Rates, One Mid-Year Population of 5,000,000

Emigration rate

62,000 emigrants

$$(62,000 / 5,000,000) \times 1,000$$

= 12.4

per 1,000 population

Immigration rate

45,000 immigrants

$$(45,000 / 5,000,000) \times 1,000$$

= 9.0

per 1,000 population

Net migration rate

$$45,000 - 62,000 = -17,000$$

$$(-17,000 / 5,000,000) \times 1,000$$

= -3.4

per 1,000 population

The State of Global Migration

304M

international migrants worldwide by mid-2024
— 3.7% of the world's population

83.4M

people living in internal displacement at end-2024 — a new record

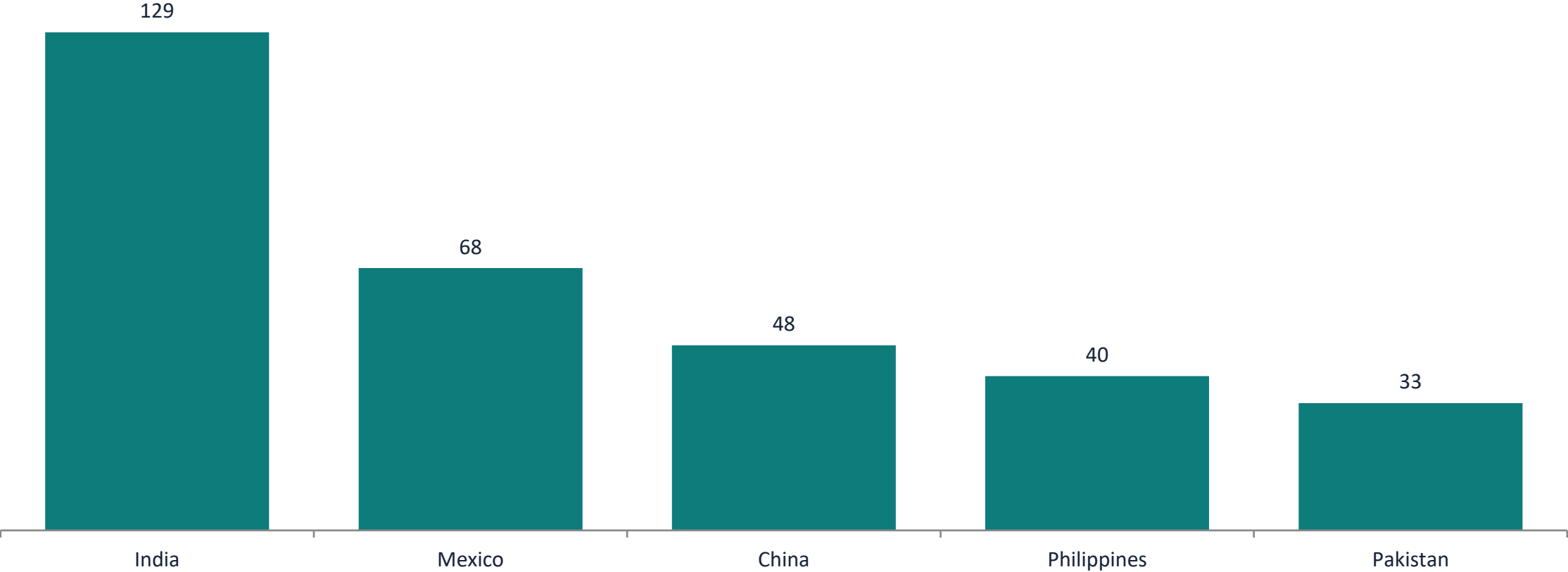
45.8M

internal displacements triggered by disasters in 2024 alone — the highest since monitoring began in 2008

Migration and displacement stocks and flows are now updated annually — but even annual figures lag fast-moving crises by months.

Source: International Organization for Migration (2024), World Migration Report 2024; Internal Displacement Monitoring Centre (2025), Global Report on Internal Displacement (GRID) 2025.

Top 5 Remittance-Receiving Countries, 2024 (US\$ Billions)



Source: World Bank (2024), Migration and Development Brief; remittances to low- and middle-income countries reached an estimated US\$685 billion in 2024.

Remittances: Migration's Largest Measurable Economic Flow

\$685B

in remittances flowed to low- and middle-income countries in 2024 — larger than FDI and ODA combined

\$64.7B

sent to Mexico in 2024 via the US–Mexico corridor, the single largest bilateral remittance corridor in the world

3.7%

of Mexico's GDP that US-origin remittances represented in 2024 — a direct demographic-to-macroeconomic link

Remittances are one of the clearest real-world consequences of migration selectivity and volume — and one of the few migration-related flows tracked in near-real time through financial data.

Source: World Bank (2024), Migration and Development Brief 41; Banco de México / Inter-American Dialogue (2025), remittance data.

Try It: Compute a Net Migration Rate

A province records 38,000 in-migrants and 52,000 out-migrants in one year. Mid-year population is 2,400,000.

NET MIGRATION (COUNT)

$$38,000 - 52,000$$

-14,000

NET MIGRATION RATE

$$(-14,000 / 2,400,000) \times 1,000$$

≈ -5.8

DISCUSS: This province is a net sender of migrants — what data would you need to know if it is also losing its most educated workers (selectivity)?

A Second Worked Net Migration Rate: Japan

Japan recorded net migration of approximately +153,357 people in 2024 against a population of roughly 123.9 million — even as births fell to a record low of 686,061.

NET MIGRATION (2024)

+153,357 people

positive

MID-YEAR POPULATION

≈ 123,900,000

123.9M

NET MIGRATION RATE

$(153,357 / 123,900,000) \times 1,000$

≈ +1.2

DISCUSS: Japan is simultaneously shrinking from natural decrease and growing slightly from net immigration — the two components can move in opposite directions in the same country.

Discuss in Pairs (5 Minutes)



1 Net migration is a residual (in minus out). Name two very different real situations that could produce the exact same net migration rate.

2 A country counts anyone staying > 1 year as a migrant; its neighbor uses > 3 months. What happens when you compare their migration rates directly?

3 Why might global totals of emigration and immigration fail to match, even though every emigrant is, by definition, someone's immigrant?



Three Real Migration Situations, One World

Assigned real, contrasting cases show migration moving for very different reasons in 2024-2025 — a destination/transit hub, a major labor corridor, and a forced-displacement crisis.

Group size: Groups of 2-3 Time: 15 min prep + 3 groups × 5 min report-back + 5-10 min synthesis discussion

COMMON GUIDING QUESTIONS — EVERY GROUP ANSWERS BOTH, REGARDLESS OF ASSIGNED CASE

1

Explore your own country. What data source best captures this flow?

2

What is one policy implication for the origin country, and one for the destination/transit country?

SEE UNDESA:
MIGRATION
STOCK

Report-back: each group presents for the allotted time, then answers both guiding questions before the synthesis discussion.

Assigned Cases

GROUP 1

Thailand — Destination & Transit

Thailand hosts at least 5.3 million non-Thai nationals (2024), including ~3.1 million regular migrant workers, mostly from Cambodia, Lao PDR, and Myanmar. Since Myanmar's 2021 coup, 2.3M+ Myanmar nationals have entered — a mix of labor migration and displacement.

ILO (2024), Thailand Migration Report 2024.

GROUP 2

Mexico → United States — Labor Corridor

About 12-13.6 million Mexican-born migrants live in the US. The corridor generated US\$64.7 billion in remittances to Mexico in 2024 — the world's largest bilateral remittance flow, equal to ~3.7% of Mexico's GDP.

Migration Policy Institute (2024); World Bank (2024).

GROUP 3

Myanmar — Forced Displacement Crisis

More than 3.5-3.6 million people are internally displaced inside Myanmar as of 2025, with UNHCR supporting over 4.2 million forcibly displaced and stateless people — driven by conflict following the 2021 military takeover.

UNHCR (2025), Myanmar Displacement Overview.



10-Minute Break

Next: migration selectivity, brain drain, and how digital data is changing what we can measure.



BLOCK 2

Selectivity, Brain Drain & Data Sources — Old and New

Who moves is never random. And how we know who moved is changing fast, from paper registers to mobile-network data.

Migration Selectivity & “Brain Drain”



Brain Drain

Migrants are rarely a random sample of the sending population — selected on age, education, skill, and ambition. When highly educated people leave in disproportionate numbers, we call it brain drain.



Fiscal & Human-Capital Cost

Origin countries lose the return on their investment in educating emigrants — a real fiscal and human-capital cost to the sending economy and its public services.



Brain Gain & Brain Circulation

Destination countries gain “brain gain” — skilled workers filling labor shortages. Some migrants later return with skills, capital, and networks (“brain circulation”), partly offsetting the loss.

Thailand: Both Destination and Transit Country

5.3M

non-Thai nationals residing in Thailand — an 8% rise on the prior report, including ~3.1M regular migrant workers

2.3M+

Myanmar nationals estimated to have moved into Thailand since the 2021 coup — a mix of labor migration and displacement

5.6%

of registered migrant workers classified as “skilled” — Thailand's inflows are dominated by low-skilled labor, the mirror image of brain drain upstream

Cambodia, Lao PDR, and Myanmar supply the overwhelming majority of Thailand's migrant workforce — making Thailand a critical case for both migration measurement and regional labor policy.

Source: International Labour Organization. (2024). Thailand Migration Report 2024. <https://www.ilo.org/sites/default/files/2024-12/thailand-migration-report-2024.pdf>

Traditional Migration Data Sources



Administrative Records

Population registers, visas, residence permits. Continuous & authoritative, but excludes unregistered/irregular migrants entirely.



Census Residence Questions

“Where did you live 1 or 5 years ago?” Covers the whole population, but only periodic (every 5-10 years) and retrospective.



Targeted Surveys

Labor-force and dedicated migration surveys give rich detail on motives & selectivity, but are costly, sample-based, and origin-country surveys can't observe emigrants who already left.

International Migrant Stock 2024: Key Facts

281M

international migrants counted in the 2024 stock estimates (UN DESA methodology)

USA

remains the single largest destination country for international migrants

61%

of international migrants live in high-income countries

Source: United Nations, DESA, Population Division. (2025). International Migrant Stock 2024: Key Facts and Figures. <https://www.un.org/development/desa/pd>

Why Traditional Sources Struggle in a Crisis



Time Lag

Censuses run once a decade; even annual surveys can take months to process and publish — far too slow for a fast-moving crisis.



Undercount of Irregular Migration

Unregistered and undocumented moves are, by design, invisible to administrative records and most registers.



Too Slow for Sudden Crises

Conflict, disaster, and displacement can move hundreds of thousands of people in days — far faster than any census cycle.

Digital Trace Data for Mobility

Anonymized, aggregated digital signals — from phones, satellites, and social platforms — can estimate population distribution and movement in near-real time, often within days rather than years.

<https://www.migrationdataportal.org/data-sources/social-media>



Mobile-Phone Data

Call detail records (CDRs) and app location pings from mobile network operators.

<https://www.migrationdataportal.org/data-sources/mobile-phone-data>



Satellite Imagery

Nighttime-lights intensity and high-resolution imagery of built-up areas and settlements.

<https://www.migrationdataportal.org/data-sources/satellite-data>



Social Media Data

Geotagged posts and platform-reported location signals aggregated at scale.

<https://www.migrationdataportal.org/data-sources/social-media>

How Call Detail Records Estimate Mobility

Mobile network operators log which tower/cell a phone connects to over time — the raw signal behind near-real-time mobility estimation.



1. Signal Collection

Mobile network operators log which tower/cell a phone connects to as the user moves, second by second.



2. Home-Location Modeling

A user's “home” is estimated as their modal (most frequent) nighttime location over a baseline period.



3. Displacement Detection

A sustained shift away from the modeled home location signals migration or displacement, often within days.

Seeing Population Change From Orbit

Satellite sensors track how brightly and how far a settlement's lights extend at night, and how built-up land cover expands — both are strong proxies for population growth and urban expansion where ground data is thin.



1. Nighttime-Lights Intensity

Rising brightness/extent over a settlement correlates with population and economic activity growth over time.



2. Built-Up Area Change

High-resolution imagery detects new construction and informal-settlement growth between census rounds.



3. Ground-Truthing

Best used validated against census or mobile data — satellite proxies alone can miss vertical (high-rise) density growth.

Example: Rwanda's Space Agency and Planet (2024) launched a national satellite-imagery program used to refine urban planning and zoning in Kigali and beyond.

Geotagged Posts as a Mobility Signal

Aggregated, anonymized location signals from social platforms can flag sudden population movement in near-real time — a complement to, not a replacement for, phone and satellite data.



1. Geotagged Activity

Posts, check-ins, and platform location settings reveal where active users are clustering or leaving from.



2. Near-Real-Time Flags

Sudden spikes or drops in activity in a location can flag a crisis or mass movement within hours.



3. Strong Selection Bias

Coverage skews toward younger, urban, platform-using populations — a major limitation for representativeness.

Dynamic Population Mapping Using Mobile-Phone Data

2014

Proceedings of the National Academy of Sciences, 111(45)

2

countries studied — Portugal and France, using mobile operator CDR data

15,888–
15,893

page range of the landmark Deville et al. study

A landmark PNAS study by Deville et al. (2014) showed that CDR data could reconstruct population distribution at fine spatial and temporal resolution, validated against census benchmarks — establishing the methodological foundation now used to estimate displacement after disasters and conflict.

Source: Deville, P. et al. (2014). Dynamic population mapping using mobile phone data. PNAS, 111(45), 15888-15893. <https://doi.org/10.1073/pnas.1408439111>

From Method to Practice: Displacement Tracking in 2025



IOM's DTM

The Displacement Tracking Matrix now formally incorporates digital trace data — mobile apps, CDRs, and social-media signals — alongside traditional field surveys to monitor crisis-driven population movement.



Flowminder's Flowkit

Flowminder Foundation operates “Flowkit,” privacy-preserving software that computes mobility indicators directly on mobile operators' servers, so raw personal data never leaves the operator.



Combined Approach

Digital signals flag where and when a movement is likely happening; field-based DTM teams verify and add context — neither source alone is sufficient.

Source: IOM Displacement Tracking Matrix. (2025). New report explores how digital trace data can strengthen displacement monitoring in crises. <https://dtm.iom.int>

When Disaster Displacement Sets a New Record

45.8M

internal displacements triggered by disasters in 2024 — the highest annual figure since monitoring began in 2008

99.5%

of 2024 disaster displacements were weather-related — many events intensified by climate change

29

countries and territories that recorded their highest-ever disaster-displacement figures in 2024, including the United States

Cyclones (Helene, Milton, Typhoon Yagi) and floods across Chad, Brazil, Afghanistan, the Philippines, and Europe drove the bulk of 2024's disaster displacement — events that mobile and satellite data can now flag within hours.

Source: Internal Displacement Monitoring Centre. (2025). Global Report on Internal Displacement (GRID) 2025. <https://www.internal-displacement.org/global-report/grid2025/>

Discuss: Ethics & Privacy of Mobility Data



- 1 Mobile-phone data can reveal displacement without consent from any individual being tracked. What safeguards would you require before using it for research or policy?
- 2 Who owns and controls this data — telecom operators, governments, or international agencies — and how might that affect what gets measured?
- 3 Digital trace data likely under-represents people without phones or network access. Which populations might be systematically invisible?

Traditional vs. Digital Data: Complements, Not Rivals

TRADITIONAL SOURCES

- + Full population coverage, legally defined categories
- + Long, comparable historical time series
- Slow: years between updates
- Blind to irregular/undocumented migration

DIGITAL TRACE DATA

- + Near-real-time — days, not years
- + Captures sudden displacement events
- Coverage biased toward phone/platform owners
- Privacy, consent & governance challenges



BLOCK 3

Urbanization in a Connected World

How the world became predominantly urban — and how the definition of “urban” itself is being rewritten in 2025.

What Counts as “Urban”?



Degree of Urbanization

The UN's 2020s methodology classifies every location worldwide as a city, town, or rural area using population size & density grids — not just national administrative boundaries.



Urban Agglomeration

The contiguous built-up area of a city, which may extend beyond a single administrative city boundary.



Urbanization Level

The percentage of a population living in urban areas at a point in time — the most commonly cited urbanization statistic.



Urbanization Rate

The pace of change in the urban share over time — distinct from, and often confused with, the urbanization level.

World Urbanization Prospects 2025

45%

of the world's 8.2 billion people already live in cities — more than double the 20% share in 1950

982M

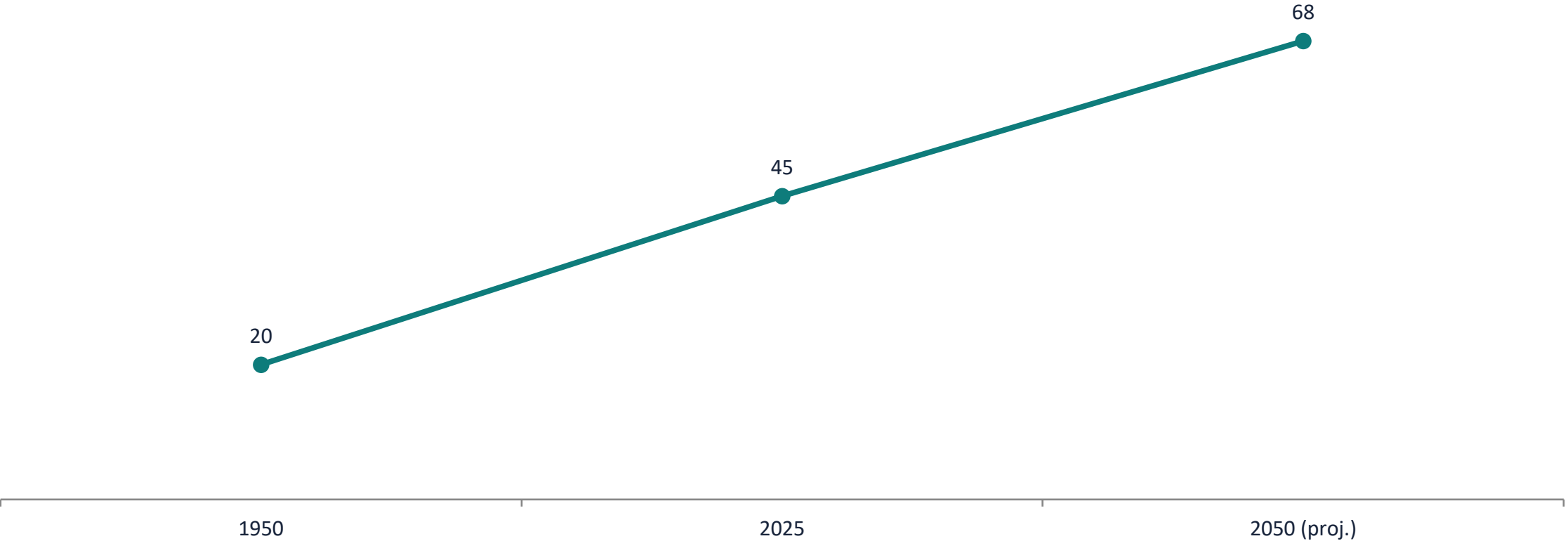
new city dwellers projected worldwide between 2025 and 2050

50,000

the new minimum population threshold for a “city” in the 2025 methodology — down from 300,000

Source: UN DESA, Population Division. (2025). World Urbanization Prospects 2025: Summary of Results. <https://population.un.org/wup>

World Urban Population Share, 1950 → 2025 → 2050 (%)



Source: UN DESA, Population Division (2025), World Urbanization Prospects 2025 (2050 figure is the medium-variant projection).

What Drives Urban Population Growth?



Rural-to-Urban Migration

People moving from rural areas to cities for work, education, and services — the classic driver of early urban transitions.



Natural Increase

Births minus deaths within the existing urban population — the dominant driver in many large cities today, including where migration has slowed.



Reclassification

Rural settlements redefined as urban as they grow, or as boundaries/definitions change (e.g., the 2025 threshold shift from 300,000 to 50,000).

Three Cities, Three Engines of Growth



Assigned real cities/countries show urban growth driven by very different combinations of migration, natural increase, reclassification, and new measurement tools in 2024-2025.

Group size: Groups of 2-3 Time: 15-20 min prep + 3 groups × 5 min report-back + synthesis discussion

COMMON GUIDING QUESTIONS — EVERY GROUP ANSWERS BOTH, REGARDLESS OF ASSIGNED CASE

1 Which of the three engines of urban growth dominates in this case, and how do you know from the evidence given?

2 What is one infrastructure or policy consequence of this growth pattern for the city or country?

Report-back: each group presents for the allotted time, then answers both guiding questions before the synthesis discussion.

Assigned Cases

GROUP 1

Lagos, Nigeria — Migration-Driven Megacity

Lagos's population reached roughly 16.5-17.2 million in 2024-2025, adding about 1.1 million people a year (≈3.3-3.75% annual growth) — reflecting strong rural-urban migration layered on high natural increase.

Statista/World Population Review (2024-2025) estimates.

GROUP 2

Thailand — Reclassification & Slower Growth

Thailand's official urban share is reported as 54.3% (World Bank, 2024) but 61.9% under alternate methodology (Statista) — the gap reflects reclassification of municipalities and administrative boundary changes, not sudden migration surges.

World Bank (2024); Statista (2024) urbanization data.

GROUP 3

Rwanda — Satellite Data for Urban Planning

In 2024, Rwanda's Space Agency partnered with Planet to roll out a national satellite-imagery program, giving government agencies daily feeds used to refine urban planning and zoning in Kigali and beyond.

Planet & Government of Rwanda (2024) program announcement.

The Migration–Urbanization–Development Nexus



1. Migration

Rural-urban and international flows concentrate people, labor, and remittance capital in cities.



2. Urbanization

Cities grow denser and larger, straining (or building) housing, transit, and service infrastructure.



3. Development

Agglomeration raises productivity — but only if governance and planning keep pace with growth.

Measuring Mobility in 2026: A Layered Toolkit



1. Census & Admin Records

The backbone — legally defined, full-coverage, long time series, but slow to update.



2. Household & Migrant Surveys

Depth on motives, selectivity, and remittances that registers can't capture.



3. Digital Trace Data

Near-real-time signals for sudden displacement, validated against ground truth.

Key Takeaways

1 Migration is a change of usual residence across a boundary; the same move is emigration for the origin and immigration for the destination.

2 Gross migration (in + out) and net migration (in – out) tell different stories — always check which one you're reading, and remember gross interchange reveals streams net figures hide.

3 Migration is selective, not random; large skilled-worker outflows are called “brain drain,” with real fiscal costs for sending countries like remittance-dependent economies.

4 Traditional sources (census, registers, surveys) remain the backbone of migration statistics, but digital trace data — mobile, satellite, and social-media — now enables near-real-time displacement estimation.

5 The world crossed 45% urban in 2025, driven by three engines (migration, natural increase, reclassification) — and how we define “urban” and “city” is itself evolving.

References

Deville, P., Linard, C., Martin, S., Gilbert, M., Stevens, F. R., Gaughan, A. E., Blondel, V. D., & Tatem, A. J. (2014). Dynamic population mapping using mobile phone data. *Proceedings of the National Academy of Sciences*, 111(45), 15888–15893. <https://doi.org/10.1073/pnas.1408439111>

Internal Displacement Monitoring Centre. (2025). Global report on internal displacement (GRID) 2025. <https://www.internal-displacement.org/global-report/grid2025/>

International Labour Organization. (2024). Thailand migration report 2024. <https://www.ilo.org/sites/default/files/2024-12/thailand-migration-report-2024.pdf>

International Organization for Migration. (2024). World migration report 2024. <https://publications.iom.int/books/world-migration-report-2024>

International Organization for Migration, Displacement Tracking Matrix. (2025). New report explores how digital trace data can strengthen displacement monitoring in crises. <https://dtm.iom.int/updates/new-report-explores-how-digital-trace-data-can-strengthen-displacement-monitoring-crises>

References (continued)

Flowminder Foundation. (n.d.). Population distribution and mobility analysis. Retrieved September 8, 2026, from <https://www.flowminder.org/what-we-do/population-distribution-and-mobility-analysis>

Migration Policy Institute. (2024). Mexican immigrants in the United States. <https://www.migrationpolicy.org/article/mexican-immigrants-united-states-2024>

Planet & Government of Rwanda. (2024). Embracing digital transformation in Africa: Planet and the Government of Rwanda launch national satellite data program. <https://www.planet.com/pulse/embracing-digital-transformation-in-africa-planet-and-the-government-of-rwanda-launch-national/>

UNHCR. (2025). Myanmar UNHCR displacement overview. <https://data.unhcr.org/en/situations/myanmar>

United Nations, Department of Economic and Social Affairs, Population Division. (2025). International migrant stock 2024: Key facts and figures. <https://www.un.org/development/desa/pd>

References (continued)

United Nations, Department of Economic and Social Affairs, Population Division. (2025). World urbanization prospects 2025: Summary of results.
https://population.un.org/wup/assets/Publications/undesa_pd_2025_wup2025_summary_of_results_final.pdf

World Bank. (2024). Migration and development brief 41: Remittances remain resilient but likely to slow.
<https://www.worldbank.org/en/topic/migrationremittancesdiasporaissues/brief/migration-and-development-brief-41>

World Bank Open Data. (2024). Urban population (% of total population) — Thailand. <https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS?locations=TH>



PRACTICE EXERCISES

Practice Exercises

18 conceptual and calculation exercises — attempt individually, then review with a classmate.

Questions 1–5: Definitions & Formulas

1. Define migration, emigration, and immigration, and explain why the same event can be described by all three terms depending on perspective.
2. Distinguish gross migration from net migration. Under what circumstances could net migration equal zero while gross migration is very large?
3. A researcher observes a migration stream from Province A to Province B, and a smaller counterstream from B to A. Define “gross interchange” and compute it if the stream is 12,000 and the counterstream is 4,500.
4. Calculation: A country has 45,000 immigrants and 61,000 emigrants in one year, with a mid-year population of 9,200,000. Compute the net migration rate.
5. Explain, in your own words, why migration rates are harder to compare across countries than birth or death rates.

Questions 6–10: Selectivity & Traditional Data

6. Define migration selectivity. Give one example of a demographic characteristic on which migrants are commonly selected.
7. Explain “brain drain” and “brain gain” using the same migration event from two countries' perspectives.
8. Using the Thailand Migration Report 2024 data from this session, explain why the low share of “skilled” migrant workers matters for both Thailand and the migrants' home countries.
9. List three traditional data sources for measuring migration and one limitation of each.
10. Why can census-based residence questions (e.g., “where did you live 5 years ago?”) undercount migration events?

Questions 11–15: Digital Data & Urbanization

11. Explain how mobile-phone call detail records (CDRs) can be used to estimate a population's “home location.” What happens when that estimate shifts?
12. Discuss one ethical concern with using mobile-network data for displacement tracking, and one safeguard that could address it.
13. How can satellite nighttime-lights data complement mobile-phone data for estimating urban population growth? Name one limitation of nighttime-lights data alone.
14. Define “degree of urbanization” as used in the UN's 2020s methodology, and explain how it differs from counting people inside official city boundaries.
15. The UN's 2025 urbanization report lowered its city population threshold from 300,000 to 50,000. Explain how this change alone could raise a country's reported urbanization level, even with no real population movement.

Questions 16–18: Synthesis

16. Name the three drivers of urban population growth and explain which one is most important in a megacity that has stopped receiving large numbers of rural migrants.
17. Using the Group Work 1 cases (Thailand, Mexico–US, Myanmar), explain why the same “net migration rate” concept applies very differently to a labor corridor versus a forced-displacement crisis.
18. A city planner has only satellite imagery and no recent census. Propose one way to combine satellite data with another data source from this session to estimate the city's current population more reliably.



Thank You

This concludes the core Fundamental Demographic Analysis content sessions — see the syllabus for the Midterm Examination (Session 9) and the final applied research paper.